



MICROSOFT GRAPH FOR AUTOMATION



LEVERAGING ENTRA MODULES, GRAPH EXPLORER, AND GRAPH X-RAY

GET-SPEAKER | WHERE-OBJECT NAME –EQ 'HARM VEENSTRA' | FORMAT-TABLE

Property	Value
Name	Harm Veenstra
Age	48 (Yes, I'm old ;-))
Job	Consultant at InSpark B.V.
MVP	2 x PowerShell (Azure)
Sessions given	5
Likes	Soccer, High BPM music, Gaming, Tech
Blog	https://powershellisfun.com
Socials	https://bsky.app/profile/harmveenstra.bsky.social https://www.linkedin.com/in/harmveenstra https://www.meetup.com/members/393325182 https://www.threads.com/@veenstra.harm https://x.com/HarmVeenstra

AGENDA

- Microsoft Graph – History, deprecation, and current Modules
- Discovering Cmdlets in Admin Portals – Graph Explorer and X-Ray Demo
- Permissions – Best practices, how to discover
- Quiz – Guess the length of the longest Graph / Entra Cmdlet 😊
- Questions (If not already answered during the session)

MICROSOFT GRAPH - HISTORY, DEPRECATION, AND CURRENT MODULES (1/2)

Select-MgProfile to switch to the Beta Endpoint

With the introduction of v2 in July 2023, the separation of Modules into normal and beta Endpoints using specific Modules (Microsoft.Graph.Users vs Microsoft.Graph.Beta.Users, etc.).

The current Module version is v2.36.1, released one month ago

These, and the Microsoft Entra Modules, replace AzureAD, AzureADPreview, and MSOnline Modules

MICROSOFT GRAPH - HISTORY, DEPRECATION, AND CURRENT MODULES (2/2)

The new PowerShell Microsoft Entra Module was released on January 23rd 2024, first Preview Edition was published on November 16th, 2023

Like the Microsoft Graph modules, it consists of Microsoft.Entra and Microsoft.Entra.beta modules for easier switching between the endpoints, have pipeline support, plus a Debug parameter to show the request URL

The current version is v1.2.0, released 3 months ago

Mapping table for old to new Cmdlets can be found [here](#)

DISCOVERING CMDLETS AND PERMISSIONS IN ADMIN PORTALS

- GRAPH EXPLORER AND X-RAY DEMO

Graph Explorer



Graph X-Ray

Featured

Extension | Merill

★★★★☆ (8) | 13,000+ Users



DISCOVERING CMDLETS AND PERMISSIONS IN ADMIN PORTALS

- GRAPH EXPLORER AND X-RAY DEMO

microsoftgraph / microsoft-graph-explorer-v4

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PowerShell code snippets not loading #4028

Open



WigF1 opened 3 weeks ago

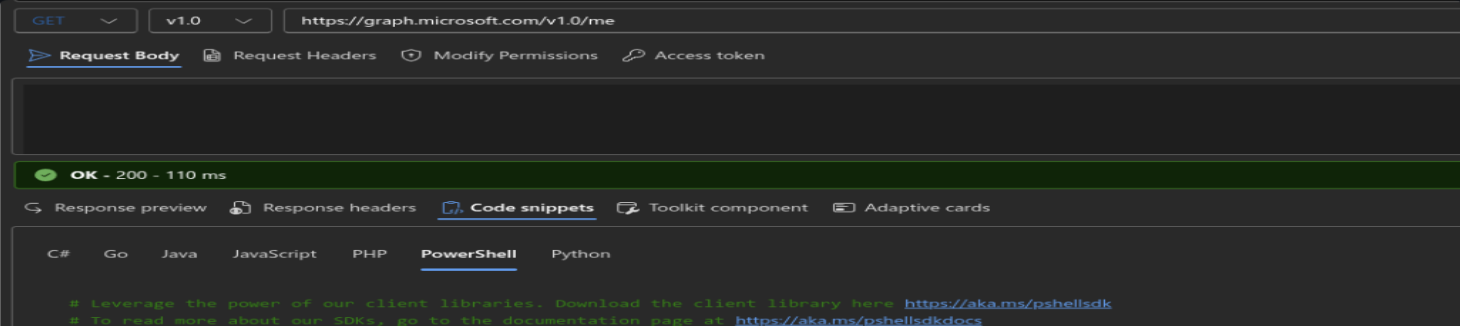
Describe the bug
PowerShell snippets are not loading (and haven't been for a month or more now)

To Reproduce
Steps to reproduce the behavior:

1. Go to Graph Explorer and perform any action

Expected behavior
PowerShell snippets should load

Screenshots
PS snippets show nothing:



DISCOVERING CMDLETS AND PERMISSIONS IN ADMIN PORTALS

- GRAPH EXPLORER AND X-RAY DEMO

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Microsoft Graph equivalent not displayed when you do an action #10

Closed



damienvanrobaeys opened on Mar 16

When you do an action, the Microsoft Graph module equivalent for an action does not appear anymore.



merill 3 weeks ago Owner

Unfortunately the back-end API no longer works. We've released a new version that falls back to show calls with Invoke-MgGraphRequest.

 merill closed this as completed 3 weeks ago

DISCOVERING CMDLETS AND PERMISSIONS IN ADMIN PORTALS

- GRAPH EXPLORER AND X-RAY DEMO



PERMISSIONS - BEST PRACTICES, HOW TO DISCOVER

Discover permissions using:

- Graph Explorer
- Microsoft Learn documentation [Graph Permissions Reference](#)
- Merril Fernando's website <https://graphpermissions.merill.net/permission>
- Find-MgGraphCommand -Uri [https://graph.microsoft.com/...](https://graph.microsoft.com/)

Connect using:

- Least permissions (Connect-MgGraph -Scopes User.Read.All, for example)
- Entra ID Application with a Certificate or Secret with limited permissions
- Disconnect-MgGraph/Disconnect-Entra when done

QUIZ

What is the length of the longest cmdlet in the graph and the ENTRA modules?

Get-

MgBetaIdentityGovernanceCatalogAccessPackageResourceAccessPackageResourceScopeAccessPackageResourceAccessPackageResourceRoleAccessPackageResourceAccessPackageResourceEnvironment with 181 characters!!

(Microsoft.Graph.Beta.Identity.Governance Powershell module)

QUESTIONS

